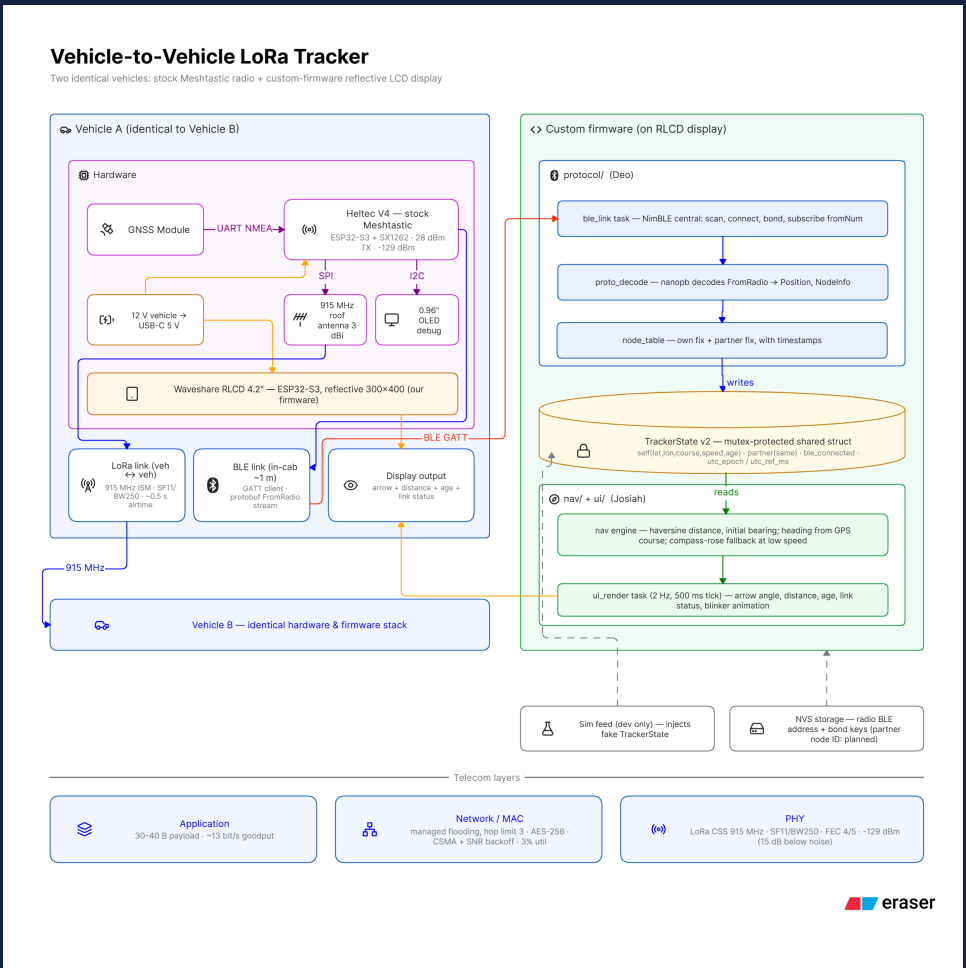
OFF-GRID VEHICLE TRACKER

TECHNICAL MANUAL, FOR ENGINEERS

Architecture, RF design, firmware, provisioning, verification, and the complete catalog of failure modes learned on real hardware. Written so an engineer who has never seen this system can operate, debug, extend, and rebuild it.



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Contents

1	System overview and the five ideas that explain everything
2	Hardware inventory and identification
3	RF design: the contract, the slot math, the modulation
4	Link budget and airtime budget
5	Security model
6	Radio configuration reference and provisioning
7	Display firmware architecture
8	Timing, liveness, and honesty semantics
9	Build, flash, and test procedures
10	Verification record
11	Failure modes and recovery (the scar catalog)
12	References

1. System overview and the five ideas

Two Heltec WiFi LoRa 32 V4 radios (callsigns SF and RG) run unmodified Meshtastic firmware 2.7.26 and exchange position, telemetry, and text over a private 915 MHz LoRa channel. Each radio serves one Waveshare ESP32-S3-RLCD-4.2 driver display running our custom firmware, connected over Bluetooth Low Energy using the Meshtastic Client API. The displays decode the radio traffic into a live navigation UI. There is no phone, tower, or internet dependency anywhere in the chain.

Five ideas explain essentially every behavior and every failure this system has ever shown. They recur throughout this manual.

IDEA 1. AGREEMENT IS THE CONNECTION. LoRa has no pairing or session. Two radios exchange data exactly when they independently agree on frequency, modulation, and encryption key. Every "total silence" failure in this project was one of those three quietly disagreeing.

IDEA 2. THE NAME IS THE FREQUENCY. Meshtastic derives the operating frequency from a hash of the primary channel name. Rename the channel and the radio retunes. The OLED LoRa screen shows the synthesized frequency and is the ground truth; configuration read-backs can lie.

IDEA 3. DECIBELS ARE RATIOS, AND LORA DECODES BELOW THE NOISE. Signal quantities are logarithmic: +3 dB is double, dBm anchors to 1 mW. At spreading factor 11 the receiver decodes down to 17.5 dB below the noise floor, because each bit is spread over 2048 chips that integrate coherently.

IDEA 4. ONLINE IS ARITHMETIC, NOT A CONNECTION. Every liveness indicator is a comparison between now and a stored last-heard timestamp. Every counter lie we chased was a timestamp problem, never a radio problem.

IDEA 5. AIRTIME IS THE BUDGET. Each packet costs roughly half a second of channel time; the firmware throttles position broadcasts above 25 percent utilization. Design every periodic packet to name the failure it prevents.

2. Hardware inventory and identification

Unit	Identity	Key facts
Radio SF	Node `!49b5ead4`, BLE MAC `8c:fd:49:b5:ea:d5`	Heltec V4: ESP32-S3, Semtech SX1262 transceiver, u-blox GNSS, OLED, 18650-class battery. Pairs with Display 1. `bluetooth.mode NO_PIN`.
Radio RG	Node `!49b5bb5c`, BLE MAC `8c:fd:49:b5:bb:5d`	Same hardware. Pairs with Display 2. `bluetooth.mode NO_PIN`.
Display 1	USB serial `A4:CB:8F:D0:7D:40`	Waveshare ESP32-S3-RLCD-4.2: reflective LCD 400x300 (sunlight readable, no backlight), runs our firmware.
Display 2	USB serial `94:A9:90:DD:14:3C`	Identical firmware; NVS-remembers RG after first scan.

Flashing trap: radios and displays share USB VID:PID 303A:1001. Radio MACs begin `8C:FD:49`. Always check the `SER=` field of the COM port before any upload; flashing a radio with display firmware erases its Meshtastic and requires factory recovery.

Enclosures: radios ride in community-printed cases; the display case is the AWSW "Symmetrical case" (Printables model 1610004, CC BY-NC 4.0), PETG, four M2.5x6 screws, 6 mm maximum or the screws press the panel glass.

3. RF design: the contract, the slot math, the modulation

3.1 The contract (Idea 1)

Layer	Our value	If mismatched
Frequency	Slot 43 = 912.875 MHz, from channel name "Group9"	Radios tuned apart; zero coupling
Modulation	LONG_FAST preset: SF11, 250 kHz bandwidth, coding rate 4:5	Chirps undecodable
Channel encryption	AES-256 pre-shared key (in the channel URL)	Packets heard, payloads dropped
Node identity	Curve25519 public keys from NodeInfo	Direct messages fail silently or NAK

3.2 Frequency slot derivation (Idea 2)

```
f_slot = f_start + BW/2 + [djb2(name) mod N_slots] * BW
N_slots = 26 MHz / 250 kHz = 104   (US band 902 to 928 MHz)
djb2: h = 5381; for each char c: h = h*33 + c; slot = h mod 104
djb2("LongFast") mod 104 = 19 → 906.875 MHz (public default, calibrates the formula)
djb2("Group9") mod 104 = 43 → 912.875 MHz (ours)
```

Consequences: two radios with identical keys but different primary channel names sit megahertz apart and never couple. The public Lubbock mesh lives on slot 19, so slot 43 carries only our traffic. The frequency is public airspace; privacy comes from the key (Section 5).

3.3 Modulation (Idea 3)

LoRa is chirp spread spectrum. Each symbol is a frequency sweep across the full 250 kHz channel; the information is the sweep's starting offset. SF11 gives 2^11 = 2048 offsets, 11 bits per symbol, and a symbol time of:

```
T_sym = 2^SF / BW = 2048 / 250000 = 8.19 ms
T_pkt = (N_preamble + 4.25 + 8 + N_payload) * T_sym ≈ 0.4 to 0.9 s per packet
```

De-chirping integrates 2048 chips coherently, which is why the SX1262 demodulates at signal-to-noise ratios down to -17.5 dB at SF11. Coding rate 4:5 adds one parity bit per four data bits for burst-error repair.

4. Link budget and airtime budget

4.1 Link budget

```
P_rx = P_tx + G_tx + G_rx - [20 log10(d) + 20 log10(f) - 147.55]
```

Term	Value	Source
Transmit power P_tx	+30 dBm (1 W)	Config, US legal max for this band (47 CFR 15.247)
Antenna gains	~0 dBi each	Stock quarter-wave whips
Sensitivity S	-131 dBm	SX1262 datasheet at SF11/250 kHz
Total budget	161 dB	P_tx - S
Design range	3 to 5 km	With 20 dB reserved for terrain, clutter, fading

Measured baseline: round-trip traceroute at desk range reports 5.75 to 7.25 dB SNR per direction, a 23.5 dB margin above the -17.5 dB demodulation floor. Rule of thumb: every 6 dB of margin roughly doubles free-space distance.

4.2 Airtime budget (Idea 5)

Meshtastic suppresses position broadcasts above roughly 25 percent channel utilization. At vehicle cadence (one ~0.5 s packet per 10 s per node) two nodes spend 10 percent. The default mesh role relays every overheard packet (hop limit 3); with exactly two nodes every relay is waste and doubles spend to ~20 percent, brushing the throttle precisely when both vehicles move. Both radios therefore run role `CLIENT_MUTE` (receive and send everything, relay nothing), measured to halve idle utilization from 9-13 percent to 5-6 percent. Revert both to `CLIENT` if a third node or repeater ever joins.

5. Security model

- **Channel traffic** (positions, broadcasts, canned texts): AES-256 with a 256-bit pre-shared key generated at channel creation and distributed only inside the provisioning URL over USB. The key never crosses the air.
- **Direct messages and position requests**: additionally encrypted per node with Curve25519 public keys exchanged via NodeInfo. A stale stored key makes direct traffic fail one-way and silently (Section 11.4).
- **What an eavesdropper on slot 43 learns**: packet headers are cleartext, so node IDs, timing, and signal levels are visible. Payloads are not. Existence and schedule leak; content and location do not.
- **Key custody**: the channel URL (embedding the key) lives in the private team repository by deliberate decision. Never publish it; rotating to a fresh channel takes minutes and orphans the old key.

6. Radio configuration reference and provisioning

Setting	Value	Rationale
`channel (primary)`	Group9, position precision 32	Private slot 43; full-resolution positions (channel default 13 truncates to a ~6 km grid)
`lora.region`	US	902-928 MHz plan
`lora modem preset`	LONG_FAST	SF11/250k/4:5, the range vs rate compromise
`lora.tx_power`	30	Legal max; margin is range
`position.position_broadcast_secs`	60	Parked cadence with fix
`position.broadcast_smart_minimum_interval_secs / _distance`	10 / 20	Moving cadence: 10 s AND 20 m
`position.gps_update_interval`	10	Must be ≤ smart interval or stale fixes re-broadcast
`telemetry.device_update_interval`	300	Liveness floor; keeps counters honest at ~0.17 percent airtime
`device.role`	CLIENT_MUTE	Two-node mesh: no relaying (Section 4.2)
`bluetooth.mode`	NO_PIN	Just-Works pairing for the display; bonded-mode radios stop advertising and become unreachable
Canned messages	21 full-word presets	Hard firmware cap: whole set is one string ≤ 200 chars (ours: 194)

Provisioning procedure (full recipe with the channel URL is in `hardware/README.md`): apply the channel URL, apply the position/telemetry/role/bluetooth settings, set the owner short name, then verify twice: read every value back over serial AND confirm the OLED LoRa screen reads 912.875. Finally power-cycle and confirm 912.875 survives; an unsaved channel write is the classic silent revert.

7. Display firmware architecture

PlatformIO project `firmware-display/`, pioarduino platform (Arduino core 3.x), roughly 2200 hand-written lines plus nanopb-generated protobufs. One-direction dataflow with a single writer and single reader around a mutex-protected store:

```
BLE task (NimBLE central)                    UI task (1 Hz-ish)
 ble_link: scan/connect/bond/subscribe/drain   tracker_store_get()
   -> proto_decode (nanopb FromRadio)            -> dead_reckon projection
   -> node_table (positions, names, batteries,   -> nav_compute (haversine,
      texts, timestamps)                            bearing, relative angle)
   -> tracker_store_set(TrackerState v5)         -> ui_render (3 pages, RLCD)
```

7.1 The contract: TrackerState (v5)

`include/tracker_state.h` is co-owned (protocol and UI sides) and versioned; changes ship as one agreed commit. v5 carries: self and partner fixes (lat/lon, course, speed m/s, age, valid), BLE link flag, GPS wall clock (epoch + millis reference), radio callsigns, latest message (seq, ms, from, text), a 4-deep received-message history, and per-radio battery levels.

7.2 BLE link (Meshtastic Client API)

GATT service `6ba1b218-...`; ToRadio write characteristic, FromRadio read (drain until empty), FromNum notify meaning "data queued". On connect: write `want_config_id`, drain the config dump (node DB, MyInfo), then drain on every notify. The radio advertises only from boot until first BLE disconnect, so the product flow is ignition-on: power radio and display together; the display fires direct connect attempts and catches the boot window. Radio address persists in NVS; a scan finding any Meshtastic service re-learns it, which is how replacement units pair automatically.

7.3 Decode and node table

Ports handled: POSITION_APP (wire speed is km/h; display stores m/s; 4.5 km/h deadband kills parked GPS jitter), NODEINFO_APP (names, keys), TELEMETRY_APP (battery), TEXT_MESSAGE_APP (raw UTF-8). Duplicate-fix detection: a bit-identical position re-broadcast does not reset the data age, so staleness stays honest. Texts from our own node number are discarded: the radio echoes sent messages over BLE, and only the receiving display should notify.

7.4 UI features worth knowing exist

Three pages (compass, status, messages; tap cycles, hold resets tracking or, on the messages page, clears history). Callsigns from NodeInfo everywhere. Link tag at 90 s freshness. Message popup for 8 s, receiver-side only. Closing/apart/holding trend from an EMA of the distance derivative. Partner battery banner at 20 percent. Hollow dart and inverted age past 30 s (Section 8). BOOT button transmits a POSITION_APP want-response packet to the partner: an on-demand position request identical to the phone app's.

8. Timing, liveness, and honesty semantics (Idea 4)

> `online(n) ⇔ t_now - t_last_heard(n) < 7200 s`

The 7200 s constant is stock firmware: `#define NUM_ONLINE_SECS (60 * 60 * 2)` in `src/mesh/NodeDB.cpp` of meshtastic/firmware, sized for an ecosystem where nodes may legitimately stay silent for hours. Our traffic contract guarantees a packet at least every 300 s, so the display layers stricter thresholds:

Indicator	Threshold	Meaning
SF<->RG OK tag	heard ≤ 90 s	Link actively fresh
Hollow dart, inverted age	age > 30 s	Position projected by dead reckoning, not measured
DATA STALE banner	age > 120 s	Stop trusting the position
Radio OLED "online"	heard ≤ 7200 s	Stock ecosystem bound

Cadence design rule: every periodic packet names the failure it prevents. Position 10 s/20 m moving (tracking staleness), position 60 s parked (presence), telemetry 300 s (counter lies), NodeInfo hours (names and keys).

9. Build, flash, and test procedures

```
# build + flash (ALWAYS pass --upload-port; check SER= MAC first)
pio run -e display-ble -t upload --upload-port COMx   # real BLE build
pio run -e display     -t upload --upload-port COMx   # SIM build (fake data, BOOT cycles scenarios)
pio device monitor --port COMx --baud 115200
pio test -e native                                    # 12 nav unit tests on the host
```

- Board flags matter: pioarduino platform, `memory_type dio_opi`, flash mode `dio`. QIO boot-loops this panel. Do not "fix" it back.
- On this workstation the native tests need MinGW on PATH: `PATH=/c/mingw64/bin:$PATH pio test -e native`.
- Every serial connection to a radio or display hard-resets it (DTR). Consequences: config writes can silently revert if the port closes mid-save (verify by read-back AND OLED), RAM-only node-table state is lost on each connect, and orphaned monitor processes hold COM ports (kill `device monitor` processes before flashing).
- Bring-up of a new display: follow the checklist in `firmware-display/README.md` (SIM proof, BLE link, partner tracking, position-request test, cased-vs-bare BLE RSSI within 2 dB).

10. Verification record

Claim	Evidence
Two-way link on Group9	Round-trip traceroutes, multiple sessions: 5.75 to 7.25 dB SNR per direction
Position accuracy at rest	Two independent fixes 13 to 45 ft apart on one table (consumer GPS scatter)
Live tracking	Display tracked a real drive on first boot; 10 s cadence proven in the Week 4 drive test
On-demand position request	CLI and display-button request both returned the partner's live fix at full precision
Messaging	Canned texts delivered both directions; popup receiver-side only (verified on glass)
Airtime claim	CLIENT_MUTE halved measured idle utilization from 9-13 to 5-6 percent
Unit tests	12/12 nav-math tests pass on native

Next planned verification: the two-vehicle drive test with metrics declared in advance (95th percentile partner-position age ≤ 15 s moving; reacquisition after separation ≤ 60 s; zero silent drops in convoy), followed by SNR-vs-distance range characterization plotted against the Section 4 prediction.

11. Failure modes and recovery (the scar catalog)

Every entry below happened on this hardware and was reproduced or root-caused before being written down.

#	Symptom	Root cause	Recovery
1	Total mutual silence; configs read back identical; 0 percent channel utilization	Frequency slot mismatch: a channel write silently reverted, radios on different slots with the same key (Ideas 1 and 2)	Check both OLED LoRa screens: Group9 must read 912.875. Re-apply the channel URL, verify, power-cycle, verify again
2	Battery ran flat; radio later boots looking healthy but config is factory or channels lost	Battery-death crash reverts NVS config (observed: one radio lost channels only, one lost everything including keys and name)	Full provisioning recipe; pre-flight rule: never run radios to empty, check 912.875 before every session
3	Screen lit, GPS fix shown, but peer hears nothing for hours and USB serial refuses to connect	Half-alive wedge after flat battery: UI runs, LoRa TX and serial frozen	Full power-cycle. Judge a radio by the OTHER radio's last-heard, never by its own screen
4	Broadcasts flow but direct messages and position requests time out in ONE direction, no NAK	Stale public key: peer's key regenerated (factory event), requester encrypts to the old key, receiver drops silently	`--reset-nodedb` on the requester, wait one minute for NodeInfo relearn, retry
5	"Online" counters lie in either direction; entries stamped months old; distance shows kilometers between adjacent radios	Stale stored timestamps and positions: no-fix radios re-broadcasting old fixes, flash-reloaded node records, clock-less boots (Idea 4)	Purge stored positions (`--remove-position`), sync clocks, reset node DBs; the 300 s heartbeat prevents recurrence
6	Config write appears to succeed, later reads show the old value	Serial port-close DTR reset rebooted the chip mid-save	Always verify by read-back AND OLED; for stubborn values use the Python API holding the port open ~20 s
7	Node table shows nameless ghost entries; freshly learned entries vanish after a USB session	Crash-corrupted records; RAM state wiped by the connect reset before the scheduled flash save	`--reset-nodedb`, then leave the radio off USB; entries repopulate from live traffic
8	Meshtastic radio erased by a firmware upload	Auto-picked COM port hit the radio (shared VID:PID)	esptool erase + factory `firmware-heltec-v4` binary at 0x0, then full provisioning; node number survives (MAC-derived)

12. References

1. Semtech SX1262 datasheet rev 2.2: sensitivity, SNR floors, PA limits
2. Meshtastic documentation: radio settings and frequency slots; mesh broadcast algorithm; device roles; telemetry; encryption (meshtastic.org/docs)
3. Meshtastic firmware source, `src/mesh/NodeDB.cpp` (`NUM_ONLINE_SECS`) (github.com/meshtastic/firmware)
4. Semtech AN1200.13: LoRa modem time-on-air
5. 47 CFR 15.247: US 902-928 MHz rules (law.cornell.edu/cfr/text/47/15.247)
6. AWSW, Symmetrical case for WaveShare ESP32-S3 4.2" RLCD, Printables 1610004, CC BY-NC 4.0
7. Project repository: hardware/README.md (provisioning recipe and gotcha log), docs/ARCHITECTURE.md, docs/RADIO_PHY_DEEP_DIVE.md, firmware-display/README.md (fleet table, bring-up checklist)